

Image Retrieval

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Abstract

Image retrieval aims to retrieve relevant images from a large dataset based on a given query image. This project explores traditional machine learning techniques for image retrieval, employing methods such as Histogram of Oriented Gradients (HoG) and Convolutional Neural Networks (CNN) for feature extraction to enhance retrieval performance. Some Image retrieval applications include e-commerce for visual product searches and medical imaging for diagnostics.

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1 Introduction

1.1 Problem Statement

The objective of image retrieval is to identify and retrieve relevant images from a dataset given a query image. This project explores feature extraction using Histogram of Oriented Gradients (HoG) and Convolutional Neural Network (CNN) to represent image content effectively. Various retrieval approaches, including classification and clustering-based techniques, are examined to improve retrieval accuracy and efficiency. The CIFAR-10 dataset is used for evaluation.

1.2 Dataset

Our dataset, which consists of 60,000 color images with a pixel size of 32×32 , is the CIFAR-10 dataset, a commonly used benchmark in computer vision. Ten mutually exclusive classes—aircraft, car, bird, cat, deer, dog, frog, horse, ship, and truck—each containing 6,000 photos, are equally dispersed among these photographs. The dataset is appropriate for both model building and evaluation because it is divided into 50,000 training photos and 10,000 test images. The photographs offer a significant challenge for classification models since they depict a broad variety of natural sceneries and object kinds with significant variations in stance, scale, backdrop, and lighting.

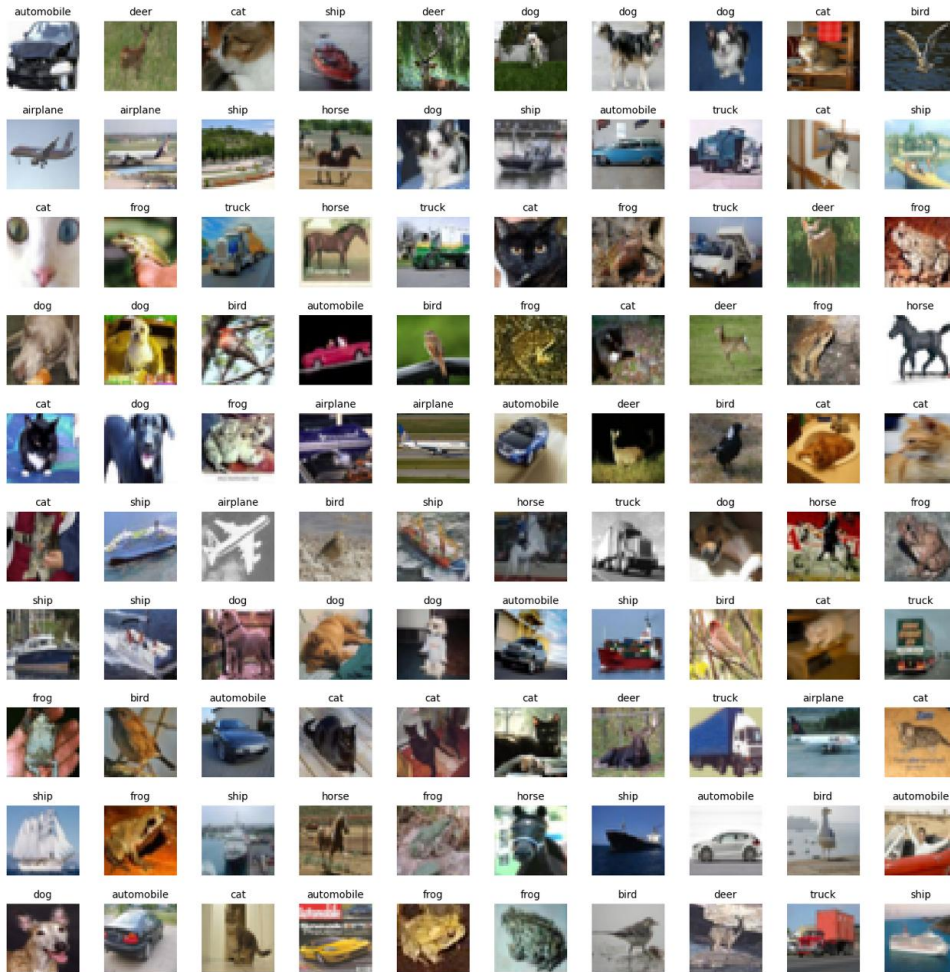


Figure 1: Random Images in Dataset

1.3 EDA

As part of our Exploratory Data Analysis (EDA), we examined the dataset to understand its structure, identify patterns, and detect dependencies among features. Based on these insights, we employed dimensionality reduction techniques such as Principal Component Analysis (PCA) and Histogram of Oriented Gradients (HoG). These methods were used to reduce feature redundancy, improve computational efficiency, and ultimately enhance the performance and accuracy of our model. To further evaluate the impact of different feature extraction and dimensionality reduction techniques, we created multiple versions of the dataset: one transformed using only PCA, another using only HoG, a combination of both PCA and HoG, as well as datasets processed through deep learning feature extractors such as ResNet and QuickNet. Additionally, we retained the original raw dataset for comparison. Each of these datasets was used to train models individually, allowing us to assess the performance of the model on each variant. This approach provided valuable insights into how various preprocessing techniques influence model behavior, accuracy, and overall effectiveness in image classification tasks.

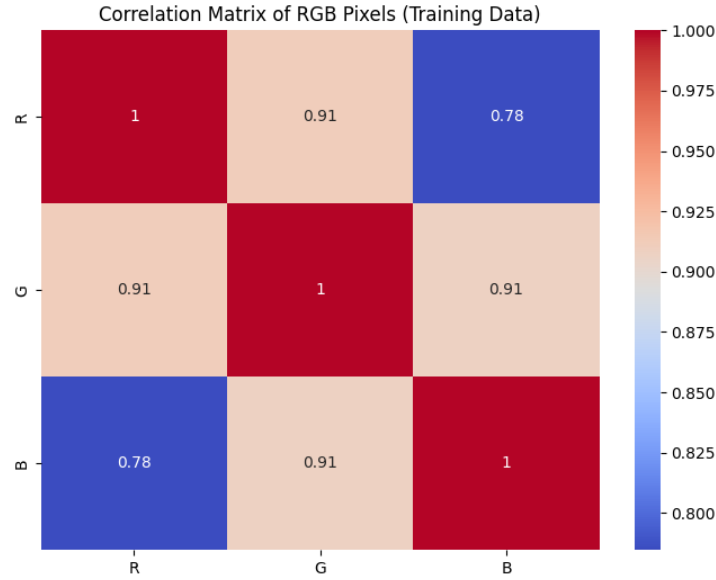


Figure 2: Correlation Matrix of Colors

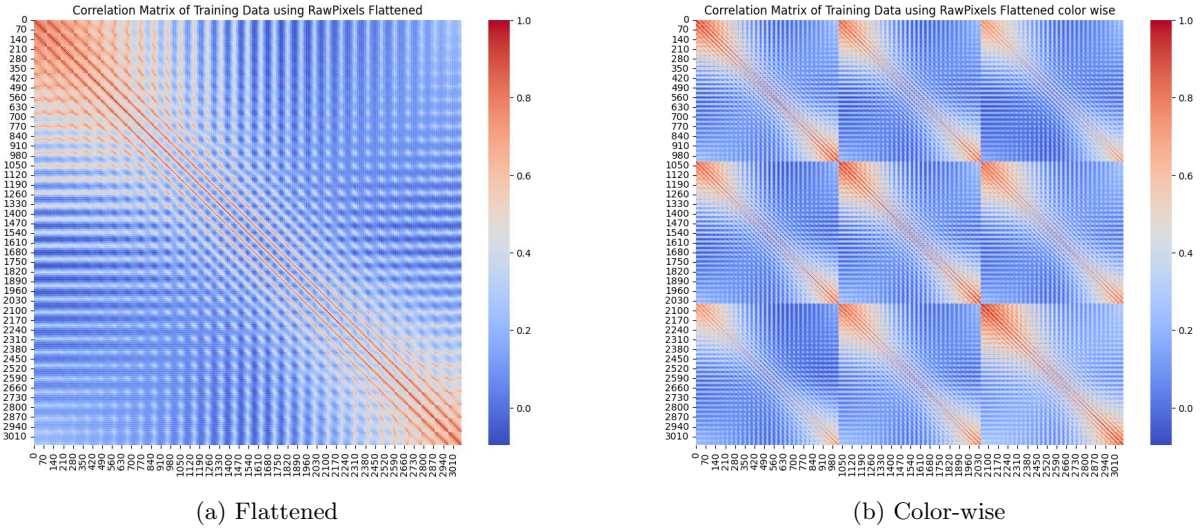


Figure 3: Correlation Matrix of Raw Features

2 Approaches Tried

2.1 Preprocessing Techniques

Before feeding the data into our models, we applied several preprocessing steps to enhance the quality and relevance of the input features. The techniques which we used are metioned below

2.1.1 PCA

As PCA aims to reduce feature correlation by transforming the original features into a new set of uncorrelated components, we applied it with a target of preserving 99% of the total variance in the dataset. This allowed us to retain most of the meaningful information while significantly reducing the dimensionality to 654 . After applying PCA, we determined the optimal number of principal components needed to reach this variance threshold and used these newly generated features for training our models.

2.1.2 HoG

We utilized the Histogram of Oriented Gradients (HoG) technique to extract structural and edge-based features from the images. HoG works by calculating the gradient orientation and magnitude within localized portions of the image, effectively capturing shape and texture information that is critical for object recognition. After applying HoG, each image was transformed into a fixed-length feature vector of 324 dimensions, significantly reducing the input size compared to raw pixel data. This compact representation emphasized essential visual patterns while filtering out irrelevant pixel-level noise. These features were then used for training and evaluating our models.



Figure 4: rhwib

2.1.3 PCA & HoG

In this hybrid approach of ours, we first applied Principal Component Analysis (PCA) to reduce the dimensionality of the raw image data while preserving 99% of the variance. This step helped eliminate noise and redundancy by transforming the highly correlated pixel features into a smaller set of uncorrelated principal components. Once the dimensionality was reduced, we applied Histogram of Oriented Gradients (HoG) on the PCA-transformed data to extract edge and texture-based features from this refined representation. The combination resulted in a compact and informative feature vector of 324 dimensions per image. This approach allowed us to work with cleaner, compressed inputs while still leveraging the spatial structure captured by HoG, offering a balanced trade-off between dimensionality and discriminative power—making it highly effective for model training.

2.1.4 HoG & PCA

In this hybrid approach of ours, we reversed the order by first applying Histogram of Oriented Gradients (HoG) directly to the original images. This allowed us to extract important edge and texture-based features, resulting in a structured feature vector of 324 dimensions per image. To further optimize computational efficiency and remove any remaining redundancy, we then applied Principal Component Analysis (PCA) on the HoG features. By retaining 99% of the variance, PCA reduced the feature vector to 170 dimensions. This two-step process preserved the essential visual information captured by HoG while significantly compressing the feature space, making it more suitable for efficient and effective model training.

2.1.5 ResNet

To harness deep learning for more abstract and informative feature representations, we utilized ResNet-50, a pre-trained convolutional neural network trained on the ImageNet dataset. Instead of using the full model for classification, we removed the final fully connected layer, effectively turning ResNet into a powerful feature extractor. Each input image was resized, normalized, and passed through the modified network, resulting in a 2048-dimensional feature vector. These deep features capture high-level patterns and semantic information, which were then used for training and evaluating our models on the CIFAR-10 dataset.

2.1.6 QuickNet

In addition to ResNet, we also explored QuickNet, a lightweight convolutional neural network architecture designed for efficiency and speed, particularly on resource-constrained devices. QuickNet offers a balance between computational cost and accuracy, making it suitable for fast feature extraction without significantly compromising performance. Similar to the ResNet approach, we removed the final classification layers and used the network to extract deep features from the images. Each image was processed to generate a compact feature vector, which retained essential visual patterns and high-level representations. These features were then used as input for our classification models to evaluate QuickNet's effectiveness on the CIFAR-10 dataset.

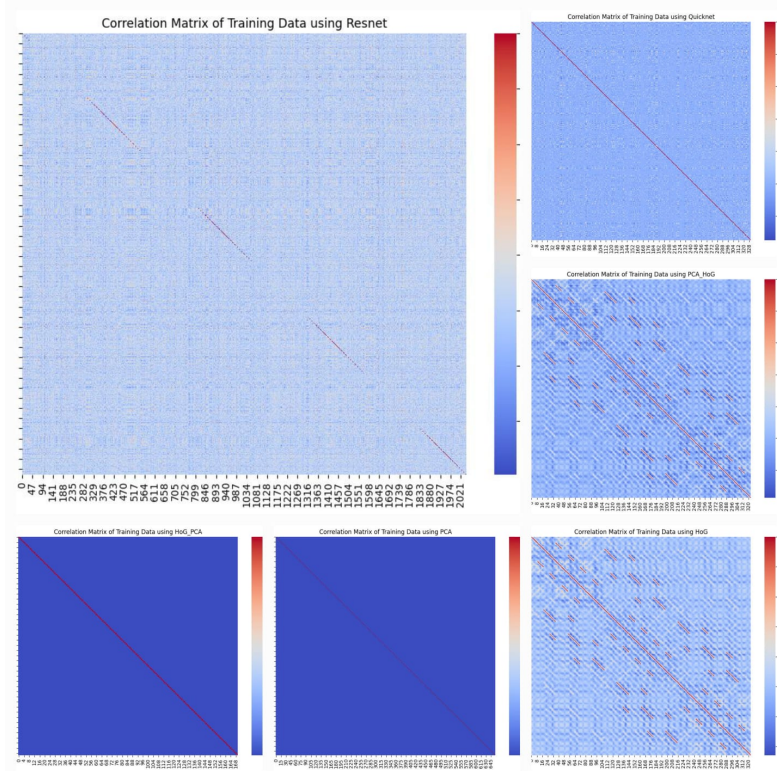


Figure 5: Correlation Matrix on Transformed Data

2.2 Classification Models

2.2.1 Decision Tree (DT) :

- We started with a Decision Tree classifier to explore the impact of various feature extraction methods. To enhance reliability, we applied 5 - Fold Cross-Validation, which helped to ensure more stable performance across different data splits.
- While raw pixels may be unsuitable for Decision Trees due to their large dimensionality, using precomputed features from deep networks or HoG enables the Decision Tree to operate on more meaningful, compact representations, boosting both performance and interpretability.

2.2.2 Gaussian Naive Bayes (GNB):

- Although the independence assumption is rarely met for images, GNB can still perform well if the features are uncorrelated (PCA components) or roughly independent.
- While Gaussian Naive Bayes may not fully capture the spatial dependencies in raw pixels, when combined with extracted features it provides a fast, simple, and interpretable classification method that serves as a strong probabilistic baseline for CIFAR-10.

2.2.3 Gaussian Mixture Model (GMM):

- GMMs model the underlying distribution of the data for each class, allowing them to capture the complex structure of image features more flexibly than simpler models like GNB.
- GMMs provide soft assignments to classes, which can be useful in uncertain or ambiguous image cases where hard labels are less reliable.
- GMMs can model classes that have multiple clusters or modes (different angles or styles of the same object), unlike models that assume unimodal distributions.

2.2.4 K - Nearest Neighbours (K-NN) :

- K-Nearest Neighbors (KNN) is a basic machine learning algorithm used for classification and regression. It makes predictions based on the labels of the 'k' nearest data points in the feature space. KNN does not require training and works well when the data is well-structured and labeled.

2.2.5 Random Forest (RF) :

- Random Forest aggregates the predictions of multiple decision trees, reducing overfitting and improving generalization compared to a single tree.
- By combining the advantages of several decision trees, Random Forest is a potent ensemble technique that produces reliable and accurate categorisation. Random Forest is good for CIFAR-10 classification since it can handle non-linear class borders and provide information about feature relevance when applied to feature representations from CNNs or PCA.

2.2.6 XG Boost (XG) :

- XGBoost includes L1 and L2 regularization, which helps control overfitting, a common issue when working with high-dimensional features.
- With numerous hyperparameters, XGBoost allows fine control over bias-variance trade-off and model complexity.

2.2.7 Logistic Regression (LR) :

- Logistic Regression outputs class probabilities rather than just hard labels, which is useful for decision-making, thresholding, or confidence estimation.
- Its probabilistic outputs and support for multi-class classification make it a valuable baseline model for evaluating feature quality and guiding further model development.

2.2.8 Supervised Centriod - Based Classification (SC) :

- Supervised Centroid-Based Classification: The model uses labeled training data to compute the centroid (mean vector) of each class, effectively summarizing each class with a representative prototype.
- Distance-Based Prediction: During testing, each input is classified by finding the nearest class centroid using Euclidean distance, making it a simple yet effective nearest-centroid classifier.
- Efficient and Interpretable: This approach is computationally efficient and easy to interpret, though it assumes that class data forms distinct, compact clusters in feature space.

2.2.9 Support Vector Machine (SVM) :

- Support Vector Machine (SVM) is a supervised learning algorithm that classifies data by finding an optimal hyperplane, maximizing the margin between different classes.
- It can handle both linear and non-linear decision boundaries using kernel functions.

2.2.10 Artificial Neural Network (ANN) :

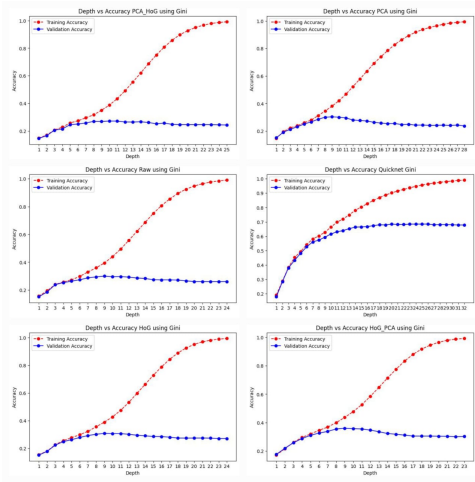
- An Artificial Neural Network is a type of machine learning model inspired by the structure and function of the human brain. It consists of layers of interconnected nodes (called neurons), where each connection has a weight that adjusts as the model learns.
- ANNs are especially useful for learning complex patterns from data and are commonly used for tasks like image classification, speech recognition, and natural language processing. By combining multiple layers and non-linear activation functions, ANNs can approximate complex decision boundaries and solve problems that traditional models struggle with.

3 Experiments & Results

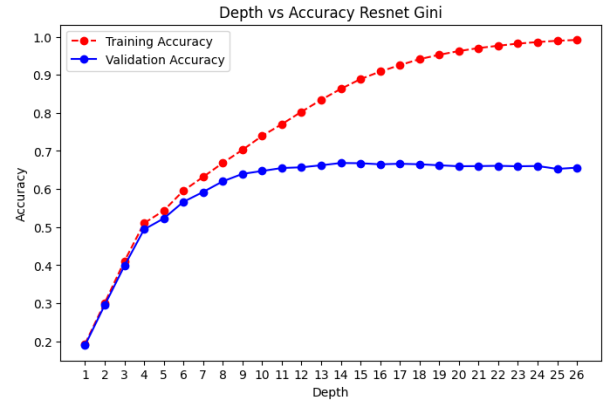
3.1 Model Performances

1. DecisionTree :

- Decision Tree classifier was implemented by mainly tuning the max depth of the tree by using the validation accuracy.
- We tried different impurity measures like Gini and entropy in which gini performed well in most of the cases, we also tried to use a generalized impurity Tsallis entropy but it was not feasible due to high computational complexity.
- We further tried to improve the performance by performing crossvalidation using Gridsearch which resulted in slight increase of the accuracy.
- We also observed that Gini impurity was producing deeper trees whereas entropy was resulting in very shallow trees.

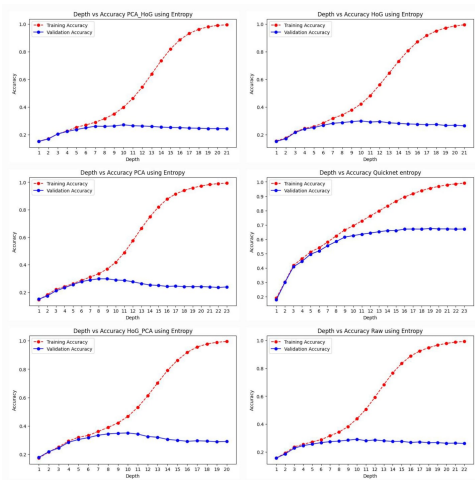


(a) Other Feature Extraction Techniques

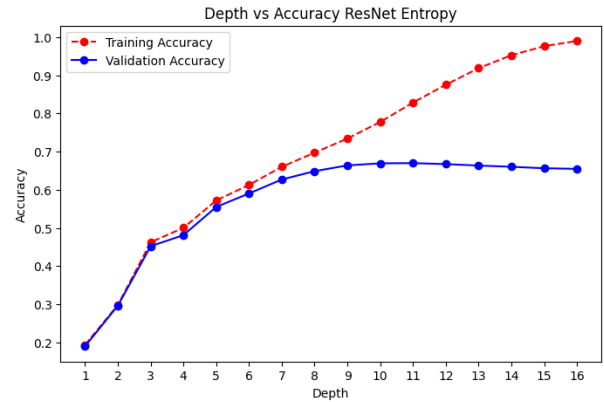


(b) Resnet

Figure 6: Train vs Validation Accuracies in Gini



(a) Other Feature Extraction Techniques



(b) Resnet

Figure 7: Train vs Validation Accuracies in Entropy

Features	Gini		Entropy	
	Cross-Val	No Cross-Val	Cross-Val	No Cross-Val
Raw Pixels	-	29.95	-	28.87
HoG	31.39	30.70	31.12	29.21
PCA	31.27	31.15	30.96	30.57
HoG & PCA	36.45	35.40	35.20	33.88
PCA & HoG	28.35	28.00	26.90	26.83
Resnet	-	66.65	-	67.34
Quicknet	-	58.01	-	57.48

Table 1: Performance of Decision Tree

2. Gaussian Naive Bayes (GNB) :

- Simple and Fast: GNB is a lightweight classifier that assumes feature independence and Gaussian distribution, making it fast to train even on high-dimensional data like image features.
- Feature-Dependent Performance: While raw pixel data may violate the independence assumption, applying GNB to preprocessed features like PCA or HoG improves its effectiveness by reducing redundancy and dimensionality.
- Good Baseline Model: Despite its simplicity, GNB provides a solid baseline for comparing more complex models, especially when using well-separated and compressed feature spaces.

Features	Accuracy (%)
Raw Pixels	29.76
HoG	45.80
PCA	28.94
HoG & PCA	47.01
PCA & HoG	43.09
ResNet	80.25
QuickNet	65.99

Table 2: Performance of Gaussian Naive Bayes

3. Gaussian Mixture Model (GMM) :

- The Gaussian Mixture Model (GMM) is a generative model that assumes each class in the dataset is made up of a mix of several Gaussian (normal) distributions. It tries to learn these distributions from the data, and during prediction, it checks which class's distribution best fits a given input. This makes GMM good at handling more complex or overlapping class patterns.
- Parameters Used and Why: n components=5: We used 5 Gaussian components for each class. This helps the model capture variations within each class (like different styles of 'cats' or 'cars' in CIFAR-10), without making things too complex or overfitting.
- covariance type = full : We chose the full covariance matrix so the model can understand how different features relate to each other—especially useful when using features like PCA and HoG where feature values aren't totally independent.
- random state = 42: This was set just to make sure the results are repeatable every time the model is run. It keeps the initialization and training consistent.

Features	Accuracy (%)
HoG	46.33
PCA	39.10
HoG & PCA	54.24
PCA & HoG	44.28

Table 3: Performance of Gaussian Mixture Model

4.K - Nearest Neighbours (K-NN) :

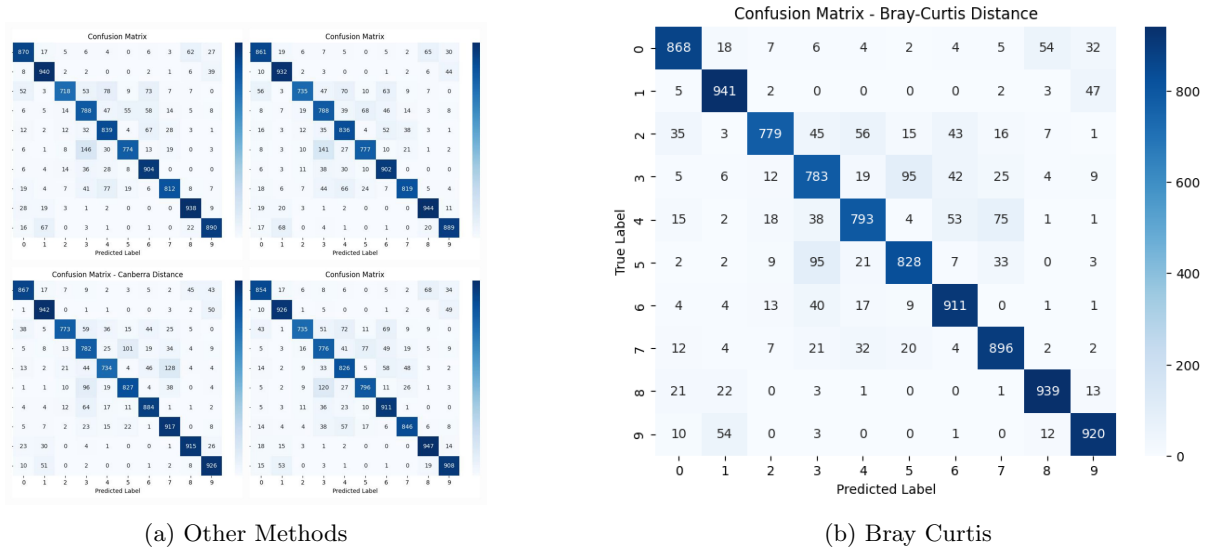


Figure 8: Confusion Matrices of KNN - Resnet

- K-Nearest Neighbors (KNN) is a simple and easy-to-use algorithm, good for starting out with the CIFAR-10 dataset. It works well with different types of features and doesn't assume anything about the data. KNN gives better results when used with features from methods like PCA, HoG, or deep learning models.
- At first, we tested our KNN model using Euclidean and Manhattan distance metrics on all types of data. Later, we added other distance metrics like Bray-Curtis and Canberra to see how they affect the model's performance.

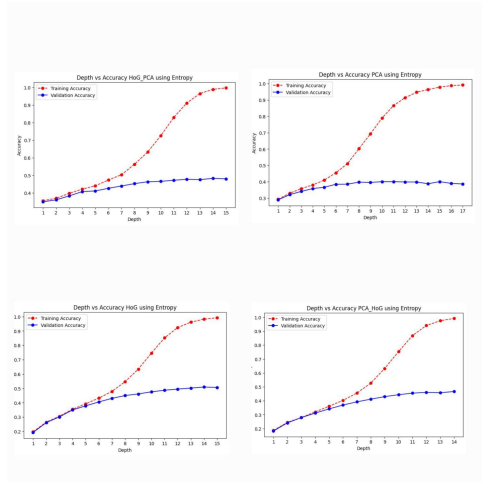
- Among all the data types, ResNet features gave the best accuracy with the KNN model. On the other hand, raw data gave the lowest accuracy, showing that unprocessed features were not very effective.
- ResNet and QuickNet transformed data perform better because they extract high-level features using deep learning. These features capture complex shapes, textures, and patterns in images. In contrast, techniques like PCA or HoG focus on basic structures or reduce data without learning from it. This makes deep model features more helpful for accurate image classification.

Features	Euclidean-C (%)	Euclidean (%)	Manhattan (%)	Canberra (%)	BrayCurtis (%)
Rawpixels	22.00	33.86	33.86	33.86	33.86
HoG	52.63	52.74	56.27	41.63	52.54
PCA	35.37	34.34	23.85	38.47	53.90
HoG & PCA	53.00	52.90	49.32	43.33	59.04
PCA & HoG	48.50	48.17	51.75	41.15	49.24
ResNet	85.25	84.83	84.73	85.67	86.58
QuickNet	72.64	72.53	72.61	73.48	74.64

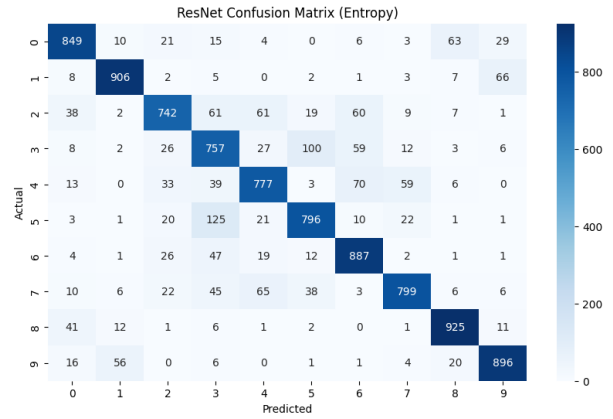
Table 4: Performance of K-NN

5. RandomForest :

- We used Random Forest from sklearn and trained it by changing the max depth to see which one gives best accuracy on validation set. We also tried to stop early when training accuracy became more than 99% to avoid overfitting. After that, we picked the model with highest validation accuracy as the final one.

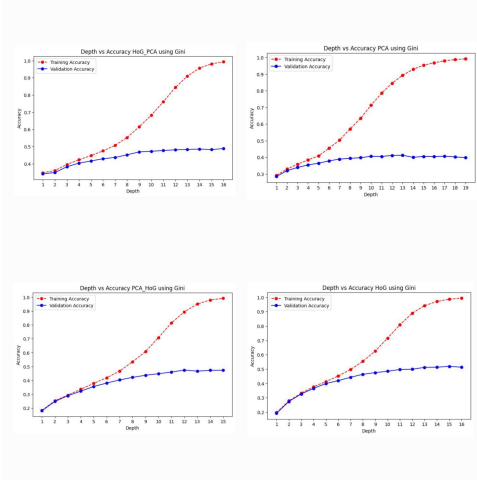


(a) Other Feature Extraction Techniques

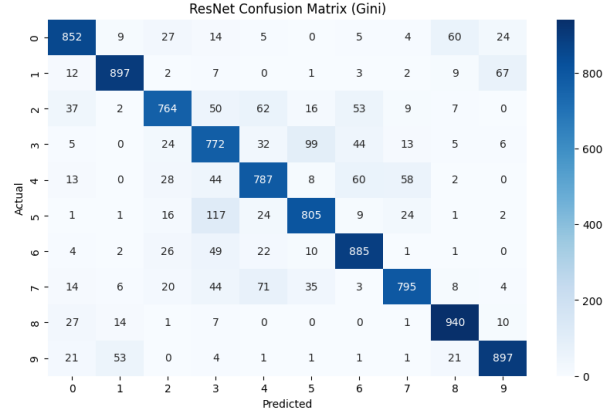


(b) Resnet

Figure 9: Train vs Validation Accuracies in Entropy



(a) Other Feature Extraction Techniques



(b) Resnet

Figure 10: Train vs Validation Accuracies in Gini

Features	Gini	Entropy
HoG	51.70	50.64
PCA	41.26	39.44
HoG & PCA	49.17	48.38
PCA & HoG	46.11	46.46
ResNet	83.94	83.34
QuickNet	73.80	73.48

Table 5: Performance of Random Forest

6. XGBoost :

- We implemented XGBoost using different hyperparameters like number of rounds, early stopping threshold and different error metrics. In comparison to Random forest the sequential processing of data in this model improves performance on the dataset.
- Used a GPU-accelerated version of XGBoost using the multi:softprob objective and monitored performance using the multi-class log loss (mlogloss) metric. To fine-tune model performance, we explored hyperparameter optimization through both GridSearchCV and Optuna, running over 30 trials.
- For HoG-PCA features, this tuning led to a notable +1.77% improvement in accuracy, indicating that XGBoost benefited from parameter selection. In contrast, ResNet-based features showed only a slight drop, suggesting that the deep features already provided a near-optimal representation for this classifier.

Features	Accuracy (%)
ResNet	86.39
HoG & PCA	57.6

Table 6: Performance of XGBoost

7. Logistic Regression :

- Logistic regression struggles with raw pixel data, likely due to high dimensionality and lack of spatial invariance.

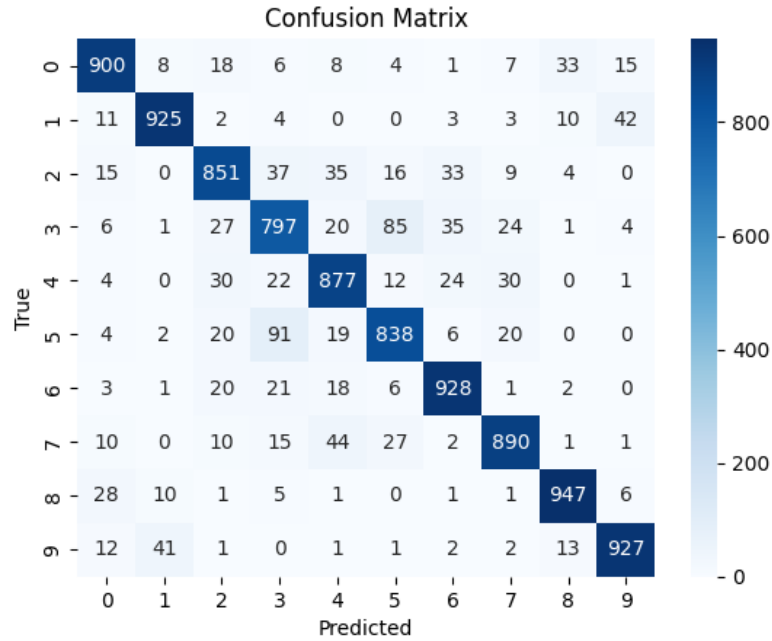


Figure 11: Confusion Matrix of Logistic Regression

Features	Accuracy (%)
Raw	38.54
HoG	54.10
PCA	39.95
HoG & PCA	53.09
PCA & HoG	49.87
ResNet	88.80
QuickNet	73.88

Table 7: Performance of Logistic Regression Model

8. Supervised Clustering :

To classify the test data, I begin by calculating the centroids of each class using the labeled training data. Next, I measure the distance between each test point and the calculated centroids. The test point is then assigned the label of the nearest centroid

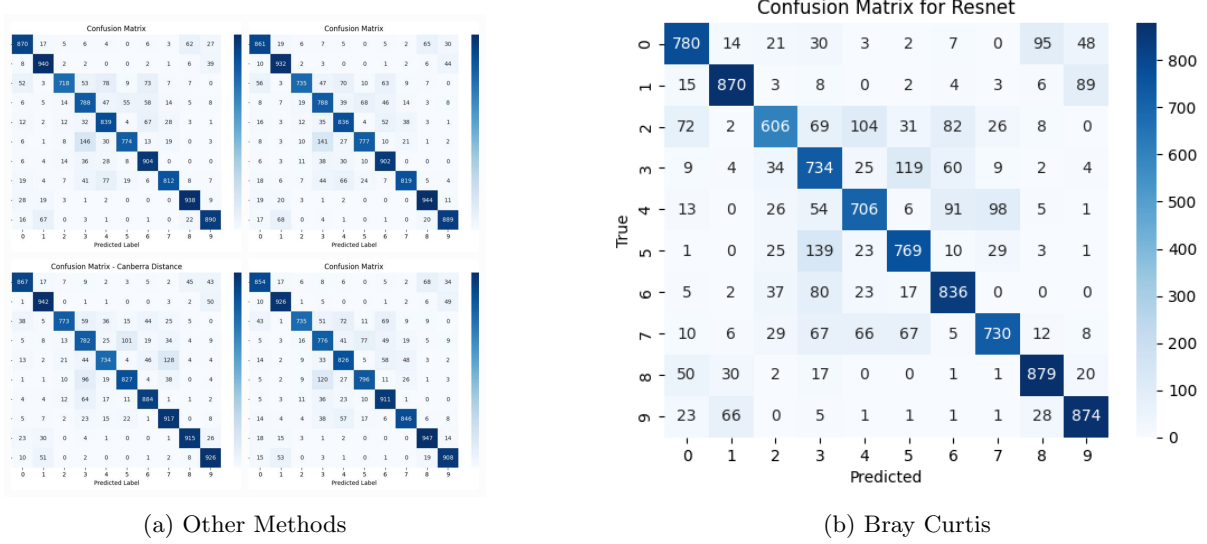


Figure 12: Confusion Matrices of KNN - Resnet

Features	Accuracy (%)
Raw	27.74
HoG	42.35
PCA	27.74
HoG & PCA	42.33
PCA & HoG	38.76
ResNet	77.84
QuickNet	68.76

Table 8: Performance of Supervised Clustering

9. SVM :

- SVM was applied across various feature sets including HoG, HoG + PCA, PCA + HoG, ResNet, and QuickNet, demonstrating its flexibility in working with both handcrafted and deep learning-based features.
- Different kernel types—Linear, Polynomial, RBF, and Sigmoid—were evaluated. The Polynomial kernel on ResNet achieved the highest accuracy of 90.28%, while Sigmoid on quicknet being the fastest with 68.75% accuracy.
- We implemented SVM from scratch on selected feature sets (HoG, HoG + PCA, PCA + HoG, QuickNet, and ResNet). This approach, although achieving a peak accuracy of 17.5% on ResNet, confirmed the core principles of SVM optimization and margin-based classification.
- These experiments affirmed that SVM, when paired with appropriate feature extraction methods, is a robust and effective classifier with strong generalization capability across various feature sets

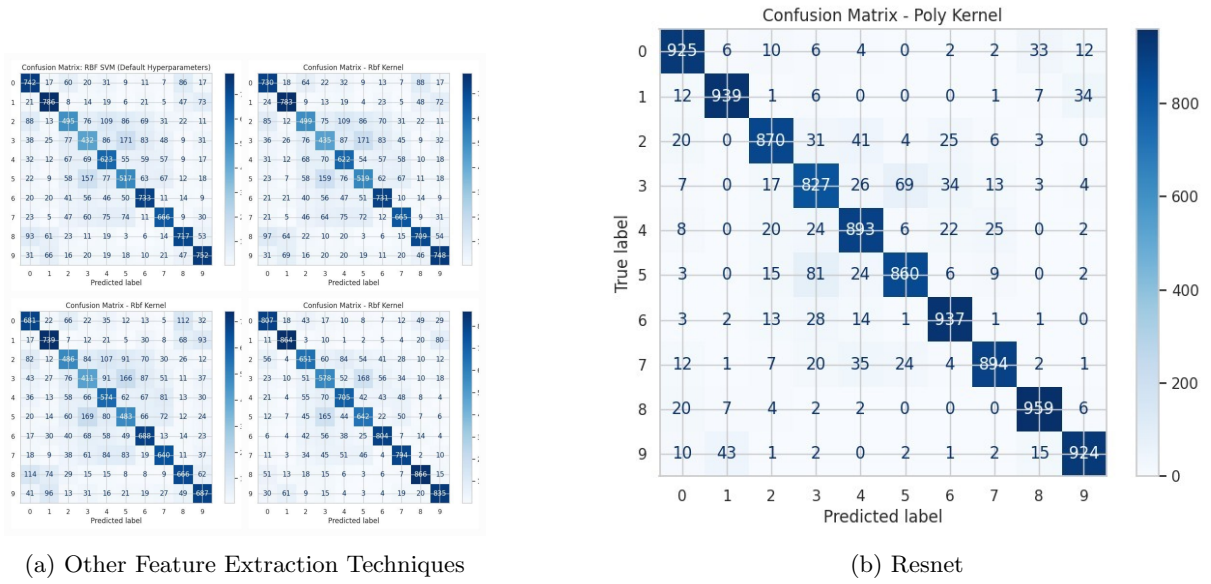


Figure 13: Confusion Matrices SVM

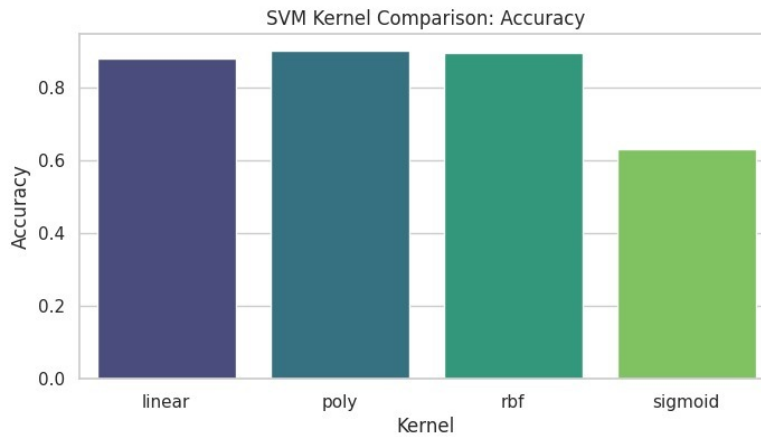


Figure 14: Accuracies of Best SVMs

Features	Linear(%)	Polynomial (%)	Radial(%)	Sigmiod (%)
HoG	54.09	63.18	64.41	17.36
HoG & PCA	53.40	62.55	64.63	40.06
PCA & HoG	50.49	58.88	60.55	15.93
ResNet	88.06	90.28	89.63	63.05
QuickNet	72.61	72.49	75.46	68.75

Table 9: Performance of SVM

10. ANN :

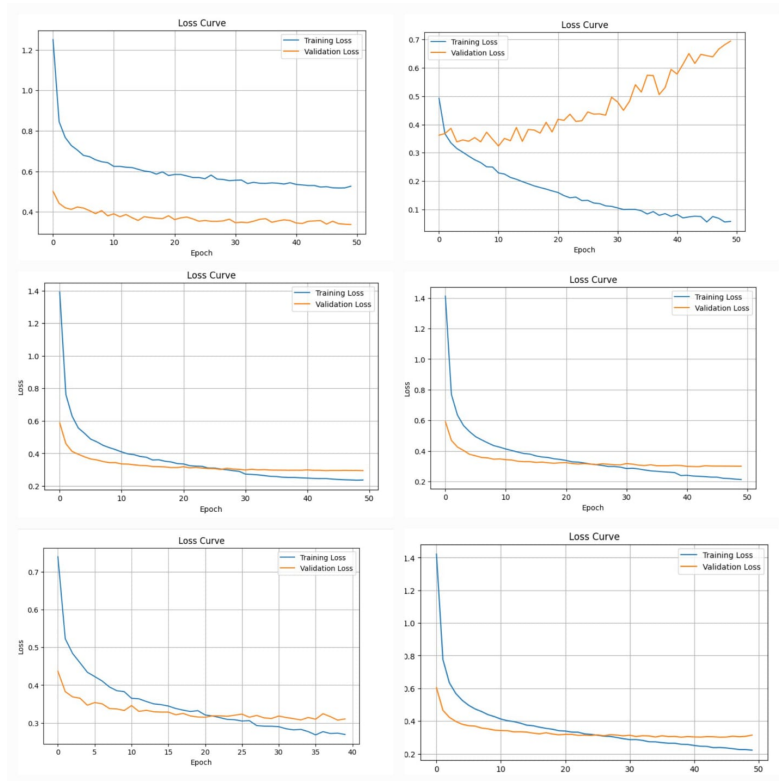


Figure 15: ANN Loss Curves for odd number of trials

- Trial 3 validation loss is above the training loss which clearly indicates overfitting. Also the ziz-zags in the validation loss plot indicates that the model is confused
- Trial 5, here the training loss is above the validation loss. Also the fluctuations in the plots has reduced significantly. These signs indicate improvement in the model because dropout rates of 0.5 was introduced.
- Trial 7 the training loss does not meet the validation loss, which is a clear sign of underfitting.
- Trial 9 plot becomes smoother because $lr = 0.0001$ was set for the Adams optimizer.
- Trial 11 is used finally

Trial	Architecture	Reg. Techniques	Val Acc (%)
1	Dense(256, ReLU) Dense(128, ReLU) Dense(10, Softmax)	None	88.63
2	Dense(256) LeakyReLU(0.01) Dense(128) LeakyReLU(0.01) Dense(10, Softmax)	None	88.77
3	Dense(128, Linear) Dense(64, ReLU) Dense(10, Softmax)	None	87.91
4	Dense(128, ReLU) Dense(10, Softmax)	None	89.42
5	Dense(512, ReLU) Dropout(0.5) Dense(256, ReLU) Dropout(0.5) Dense(10, Softmax)	Dropout(0.5)	89.30
6	Dense(256, ReLU) Dropout(0.5) Dense(128, ReLU) Dropout(0.5) Dense(10, Softmax)	Dropout(0.5)	89.64
7	Dense(128, ReLU) Dropout(0.5) Dense(64, ReLU) Dropout(0.5) Dense(10, Softmax)	Dropout(0.5)	89.25
8	Dense(256) + BN ReLU + Dropout(0.5) Dense(128) + BN ReLU + Dropout(0.5) Dense(10, Softmax)	BatchNorm Dropout	90.26
9	Dense(256, ReLU) Dropout(0.5) Dense(128, ReLU) Dropout(0.5) Dense(10, Softmax)	Dropout(0.5) learning rate 0.0001	90.06
10	Same as Trial 9	Dropout(0.5) EarlyStopping	90.29
11	Same as Trial 9	Dropout(0.5) EarlyStopping ReduceLROnPlateau	90.27

Table 10: Summary of ANN using Resnet Features

3.2 Model Comparisions

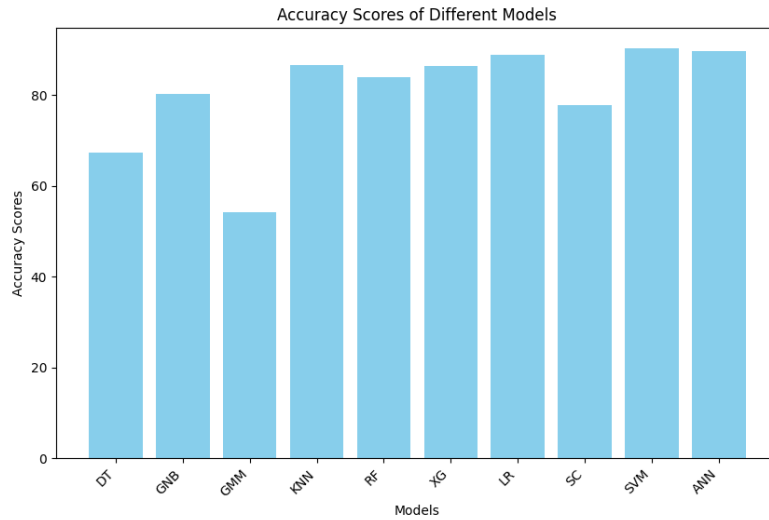


Figure 16: Accuracies

4 Summary

- Among all the models ANN was the best along with resnet features with an accuracy of 89.67 %.
- From the confusion matrix of that model we can infer that many misclassifications are between class 5(dog) & 3(cat) because their features are quite similar considering the dimensions of the images in the dataset.
- We can see that CNN based features perform very well on image based classification

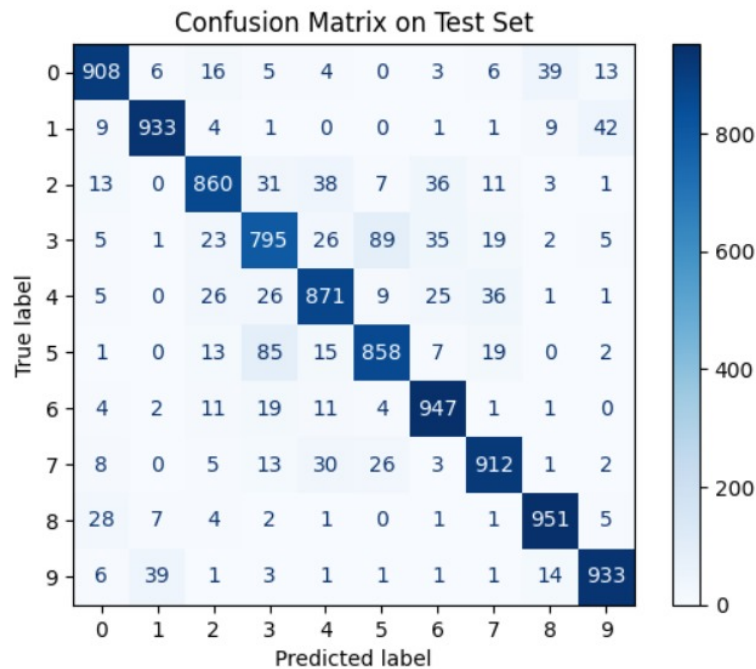


Figure 17: ANN confusion Matrix

5 Citations

- **Scikit-learn website** - <https://scikit-learn.org/stable/>
- **Stack Overflow** - <https://stackoverflow.blog/machine-learning/>
- **KNN** - <https://globaljournals.org/item/3888-analysis-of-distance-measures-in-content-based-image-retrieval>
- **XGBoost** - <https://youtu.be/GrJP9FLV3FE?si=SMZQtfrEpvGig4A5>
- **Logistic Regression** - <https://youtube.com/playlist?list=PLblh5JKOoLUKxzEP5HA2d-Li7IjKHfXSesi=KuSygPKYxv6>
- **Random Forests** - https://youtu.be/J4Wdy0Wc_xQ?si=1IzPuh43HxeT3200

6 Contribution of each member

- **Vara Prasad Reddy** : Performed feature extraction excluding CNN based feature extraction, implemented DecisionTree, XGBoost, report making, webdemo creation.
- **Kodakandla Manideep** : Performed feature extraction excluding Quicknet based feature extraction, EDA, implemented DecisionTree, Logistic regression, Supervised centroid based classification, report making.
- **Neeraj** : Implemented GNB, GMM, ANN , creation of webpage, spotlight video.
- **Kalkar Tejas Prasad** : Implemented SVM, KNN, performed feature extraction for Quicknet features, Supervised centroid based classification, made the presentation, made MOM
- **Priyanshu** : Implemented KNN, SVM, made the presentation, performed feature extraction for Quicknet features, Supervised centroid based classification, made MOM
- **Yadav Karan Shubashchandra** : Implemented Random Forest, MOM making, Git repo maintenance